

SOLUTIONS - Exercise - Week 8: MATH1061/7861

Let $f : \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $f(x) = 2x^2 - 1$.

Let $g : \mathbb{Z} \rightarrow \mathbb{Z}$ be defined by $g(y) = y + 2$.

- (1) **(1 mark)** What is the image of $\mathbb{Z}_{\geq 2}$ under g ?

The image is $\mathbb{Z}_{\geq 4}$.

MARKING:

1 mark for correct answer.

- (2) **(1 mark)** Determine $f^{-1}(7)$.

The pre-image will be solutions to $2x^2 - 1 = 7$, which are equivalent to solutions to $x^2 = 4$. Thus, the pre-image is $f^{-1}(7) = \{-2, 2\}$.

MARKING:

1 mark for correct answer, including set notation.

- (3) **(2 marks)** Is f one to one? Justify your answer.

The function f is not one to one. For example, notice that $f(2) = 7 = f(-2)$.

MARKING:

1 mark for correctly identifying the function is not one to one.

1 mark for providing a valid counter-example.

- (4) **(2 marks)** Is f onto? Justify your answer.

The function f is not onto. For example, there is no integer x such that $f(x) = 3$. If there was such an integer x , then we would have $2x^2 - 1 = 3$ and so $x^2 = 2$, which is only satisfied by $\sqrt{2}$ and $-\sqrt{2}$ but these are not integers.

MARKING:

1 mark for correctly identifying the function is not onto.

1 mark for providing a valid counter-example.

- (5) **(1 mark)** Determine the composition $f \circ g$.

The composition $f \circ g$ is the function mapping integers x to $2(x + 2)^2 - 1$. Equivalently, it maps integers x to $2x^2 + 8x + 7$.

MARKING:

1 mark for correct answer.

- (6) **(1 mark)** Determine the composition $g \circ f$.

The composition $g \circ f$ is the function mapping integers x to $(2x^2 - 1) + 2 = 2x^2 + 1$.

MARKING:

1 mark for correct answer.

- (7) **(2 marks)** Is $g \circ g$ is bijective? Justify your answer.

The function $g \circ g$ is bijective. First, to simplify our working, notice that $(g \circ g)(x) = (x + 2) + 2 = x + 4$.

To show $g \circ g$ is injective, suppose that $(g \circ g)(x) = (g \circ g)(y)$ for some integers x and y . Then $x + 4 = y + 4$, so $x = y$ and thus $g \circ g$ is injective.

To show $g \circ g$ is surjective, take any $y \in \mathbb{Z}$. Then $(g \circ g)(y - 4) = (y - 4) + 4 = y$, and hence $g \circ g$ is surjective.

As $g \circ g$ is both injective and surjective, it is bijective.

MARKING:

1 mark for injectivity.

1 mark for surjectivity.